

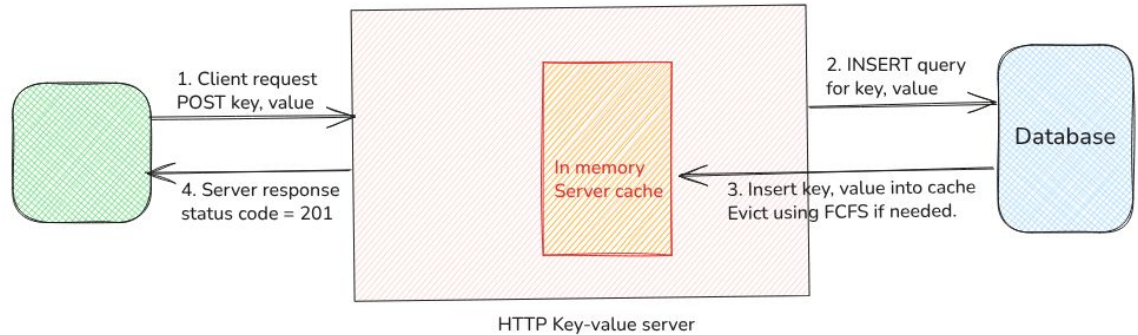
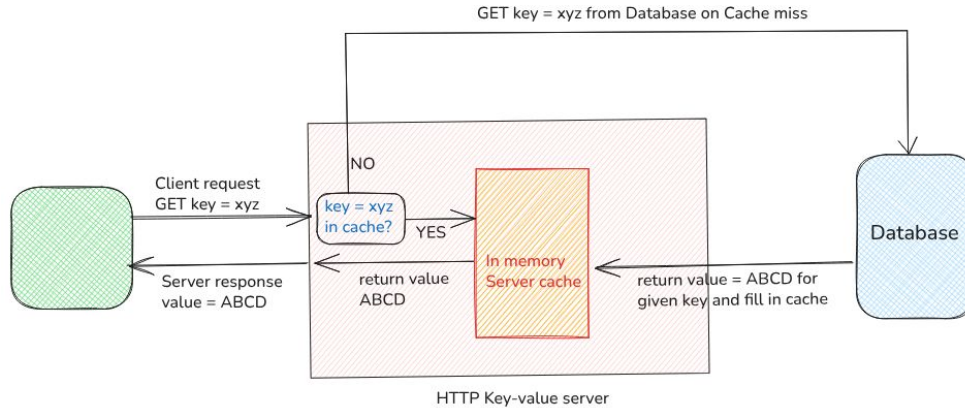
CS744 Project Presentation

Key-Value Server Performance Analysis

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Architecture Diagram



Development Process

- 8 total CPU cores in machine.
- Used [cpp-httplib](#) to write a multithreaded HTTP server pinned to 2 CPU cores.
- 1 master thread to accept incoming connection and 8 worker threads to process incoming requests.
- PostgreSQL as database pinned to 3 CPU cores and used [libpq](#) library in server to interact with the database.
- `std::unordered_map<string, string>` as in-memory cache.
- C++ STL based load generator pinned to 3 CPU cores.
- Mutex for locking the cache entirely for any operation (coarse-locking).

[GitHub Link](#)

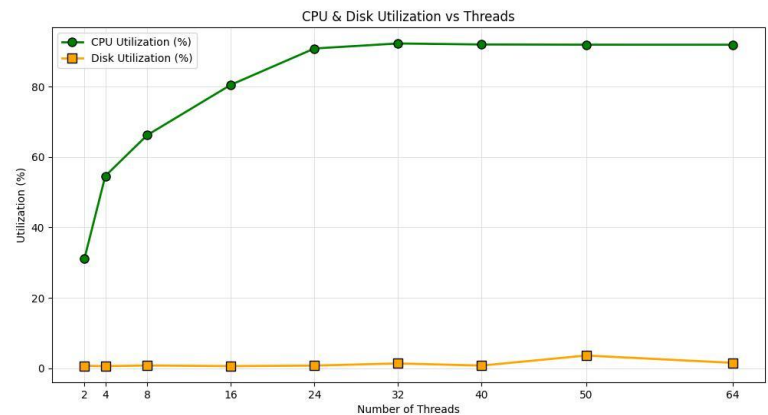
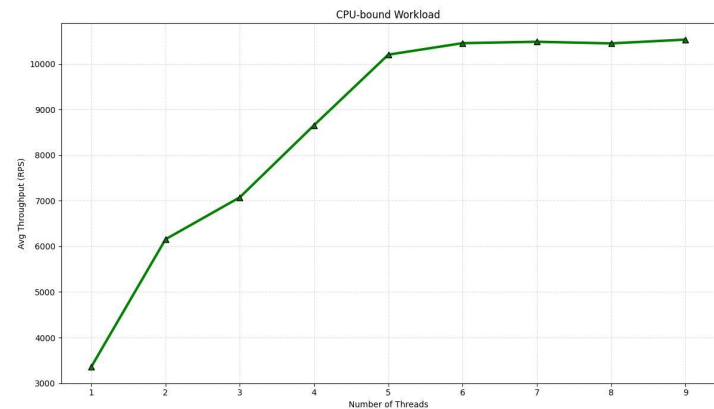
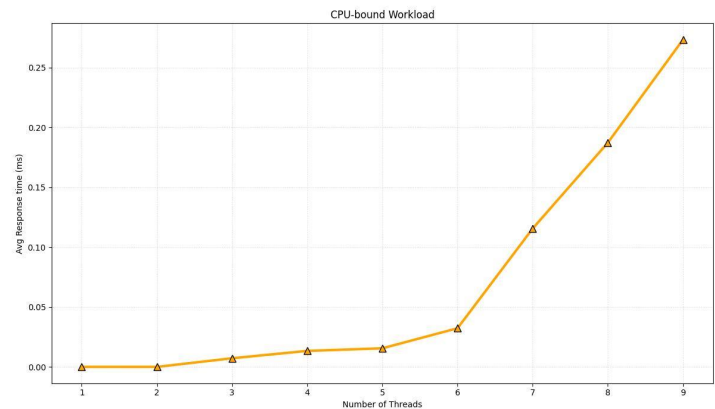
Load Generator

- Two load generator programs to simulate CPU bound and I/O bound workload.
 - For CPU bound load testing, each client (thread) sends GET requests to fetch the value of one of 10 keys. Cache hit rate > 99%.
 - For I/O bound load testing, each client (thread) sends POST requests to insert key-value pairs into the database.
- Closed loop load generator with each thread simulating requests from one client.
- Ensured CPU utilisation of load generator cores not high to maintain high throughput of outgoing requests.

Load Test Setup

- Load generator program pinned to 3 cores and server pinned to 2 other cores in the same machine. Database pinned to the remaining 3 cores.
- Total 10 experiments for CPU bound load testing.
 - Thread count ranging from 1 to 9.
 - Duration of each experiment - 5 minutes.
- Total 10 experiments for I/O bound load testing.
 - Thread count ranging from 2 to 64 in steps.
 - Duration of each experiment - 5 minutes.
- Measured parameters:
 - Average throughput in requests/second.
 - Avg response time for a client request.
 - CPU and Disk utilisation.

CPU-bound Load Test



Disk-bound Load Test

